

PRODUCT SHOWCASE

Flow

Long-term work made visible and doable day by day, based on real capacity, with a clear monthly budget.

A mobile-first PWA designed, specified and shipped in 3 weeks, with AI coding agents doing the implementation and me as Product Owner.

Source code is private. This deck showcases the product, architecture and key decisions. Fictional demo data.

THE PROBLEM

Deadlines months away get pushed until they collide.

To-do lists hide the long term

A thesis or a side project never shows up as "today's task" until it is already late.

Plans ignore real capacity

Classes, client calls, trips across time zones: the free time that actually exists is never computed.

Money is scattered

Expenses in GBP, EUR, USD across countries; no clear view of the month, of envelopes or of a trip budget.

Target user: **one demanding user** (me: a master's student who freelances and travels continuously). Every feature is tested on a real phone, in real use, every day.

Projects as stocks of minutes, spread over real free time.

1 · Capacity, not a list

Each day = window – calendar blocks – trip buffers – planned tasks. Projects are spread toward their deadline in 30-min chunks or full sessions.

2 · Debt with a date

What doesn't fit becomes visible debt, with the date the project would slip. One tap replans the week.

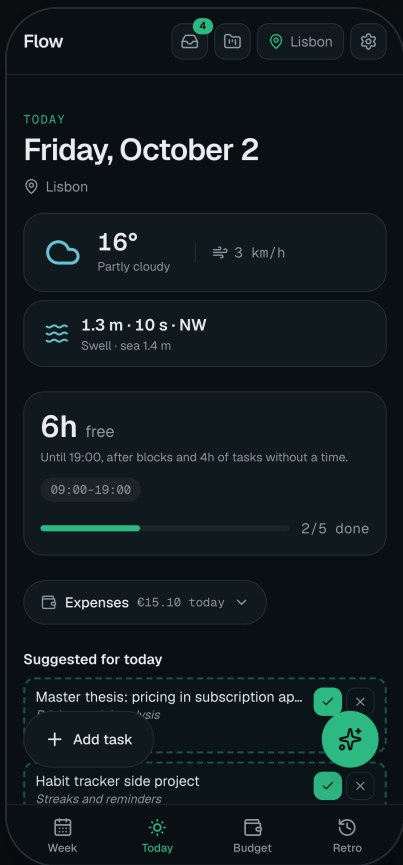
3 · Budget in two gestures

Type an amount, tap a category: saved, in the right currency and country. Envelopes, pots and an end-of-month projection.

4 · The AI proposes, I validate

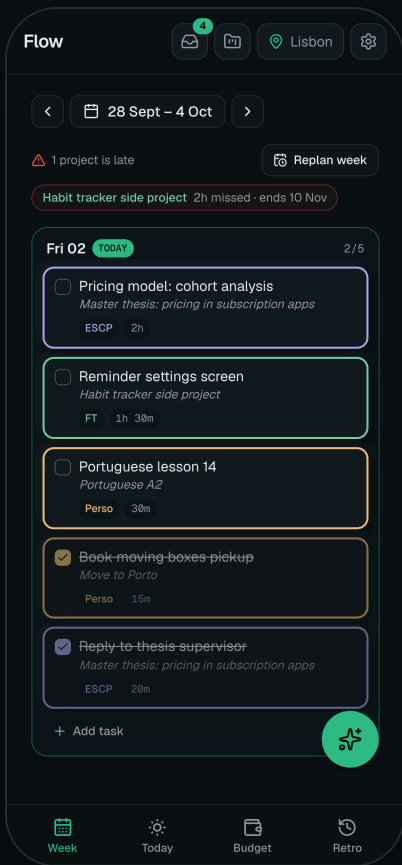
Chat, booking emails and shortcuts only create **proposals** in an Inbox. Nothing is written without a tap.

Today, Week, Retro



Today

Free time left, stay, weather and swell, suggested sessions.



Week

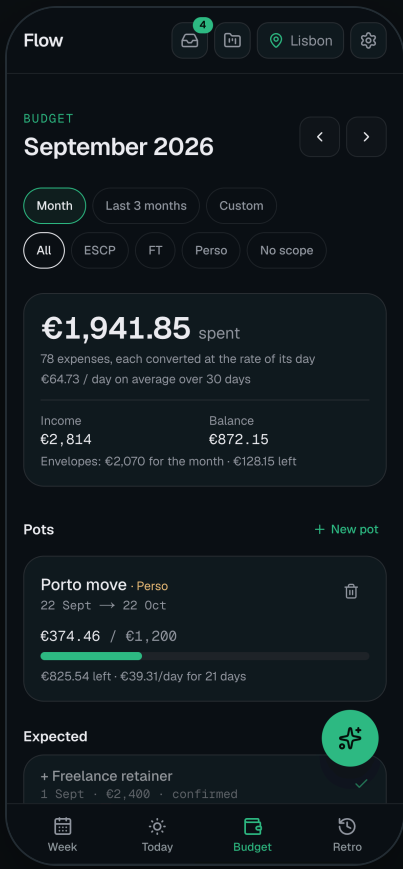
Capacity-aware plan, late projects and slip dates.



Retro

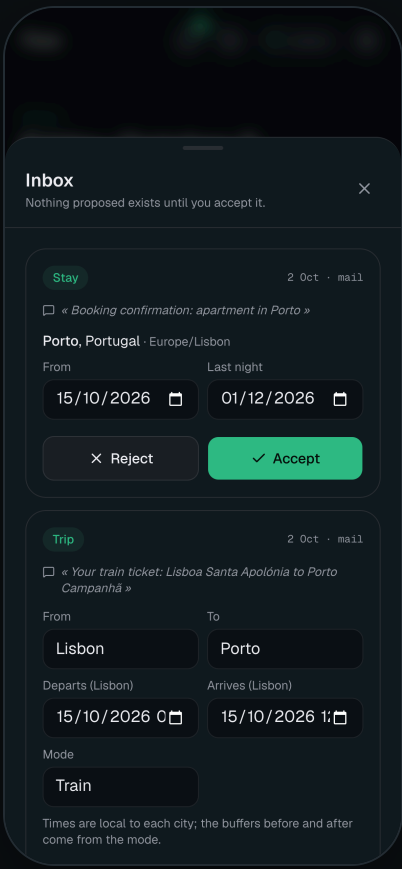
Completion, streaks and focus over the year.

Budget, Inbox, Assistant



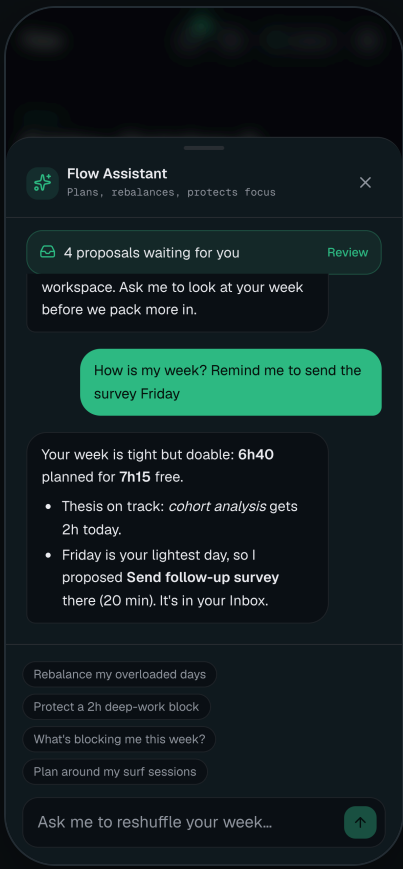
Budget

Multi-currency at the day's rate, envelopes, pots.



Inbox

Booking emails become editable proposals.



Assistant

Knows the week; proposes, never writes.

ARCHITECTURE

One Next.js app, server-side AI, owner-only data.

IPHONE · PWA

Next.js UI

Today · Week · Budget · Retro · Inbox · Chat

Service worker

cached screens + last reads

Local outbox

expense, task, check: client ids, no duplicates

iOS Shortcut

Action button, personal token

VERCEL · FRA1

Server Actions + Queries

modules/*/actions.ts · queries.ts, no DB call in components

core/ · pure engine

capacity, allocation, time zones, money: unit-tested

lib/ai adapter → Gemini

tools only **propose** (captures)

API routes + cron

/api/chat · /api/capture/email · /api/shortcut · ICS sync · FX

DATA AND OUTSIDE

Supabase Postgres

owner RLS on every table, 20 migrations, RLS check script

captures

pending → accepted / rejected (atomic)

Gmail + Apps Script

JSON-LD first, one AI call as fallback

ICS · Currency API · Open-Meteo

free, keyless, server-side only

6 of 31 documented decisions

AI

The AI proposes, I validate

Assistant tools only create Inbox proposals; atomic accept. **Trade-off:** one more tap, zero silent writes.

PLANNING

Capacity-based allocation

Pure engine spreads minutes to the nearest deadline; overflow = dated debt. **Trade-off:** projects need hour estimates.

CAPTURE

Booking emails without an agent

schema.org first, one AI call only if needed, rules in code, idempotent. **Trade-off:** less magic, deterministic and free.

MONEY

Rate frozen per day

Integer minor units, EUR value stored at the day's rate; date $\neq$ stay. **Trade-off:** more logic per row, stable history.

OFFLINE

Offline only where it matters

Hand-written service worker; 3 queued writes with client ids. **Trade-off:** last-write-wins, small surface.

PROCESS

Real environments, solo

Branch + preview per iteration, dev/prod split, merge by the PO only. **Trade-off:** more process, safe daily shipping.

METHOD

AI agents build. I own the product.

- **Product Owner role:** vision, roadmap cut into iterations, acceptance criteria, testing every build on my iPhone, go/no-go for production.
- **Claude Code as the dev team**, framed by a written contract (**CLAUDE.md**): architecture invariants, forbidden actions (secrets, `main`, unapproved migrations, out-of-scope refactors).
- **Rituals as commands:** `/recap` → `/ship` (checks, push, preview link, delivery report) → `/handoff`; `/goprod` only on my call.
- **Decision log:** 3 lines max, never revisited without a new entry. A "limits" log instead of silent scope creep.

Definition of Done, every iteration

- ✓ typecheck · lint · tests
- ✓ RLS checked on every table
- ✓ works one-handed on iPhone
- ✓ delivery report with known limits
- ✓ acceptance criteria validated by me on the live preview

3 weeks, shipped daily.

126

commits

25

pull requests

22

iterations with
acceptance criteria

31

documented
decisions

20

DB migrations

Unified retro

A weekly score based on long-term progress; time and money debt in one view.

Journaling

Daily notes feeding the retro and the assistant's context.

Anticipation agent

Flags conflicts (a deadline vs. a trip) before they happen, still as proposals.